

Problem

Express the Fourier series

$$f(t) = 10 + \sum_{n=1}^{\infty} \frac{4}{n^2 + 1} \cos 10nt + \frac{1}{n^3} \sin 10nt$$

(a) in a cosine and angle form,

(b) in a sine and angle form.

Step-by-step solution

Step 1 of 2

$$(A) \quad D \cos \omega t + E \sin \omega t = A \cos(\omega t - \theta)$$

Where,

$$A = \sqrt{D^2 + E^2}, \quad \theta = \tan^{-1} \frac{E}{D}$$

$$A = \sqrt{\frac{16}{(n^2 + 1)^2} + \frac{1}{n^6}} = \tan^{-1} \left(\frac{(n^2 - 1)}{4n^3} \right)$$

$$f(t) = 10 + \sum_{n=1}^{\infty} \sqrt{\frac{16}{(n^2 + 1)^2} + \frac{1}{n^2}} \cos \left(10nt - \tan^{-1} \frac{n^2 + 1}{4n^3} \right)$$

Step 2 of 2

$$(B) \quad D \cos \omega t + E \sin \omega t = A \sin(\omega t + \theta)$$

Where,

$$A = \sqrt{D^2 + E^2}, \quad \theta = \tan^{-1} \left(\frac{D}{E} \right)$$

$$f(t) = 10 + \sum_{n=1}^{\infty} \sqrt{\frac{16}{(n^2 + 1)^2} + \frac{1}{n^2}} \sin \left(10nt + \tan^{-1} \frac{4n^3}{n^2 + 1} \right)$$